The empirical product formula for OEIS A140940, proved

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Abstract

OEIS A140940 counts 8×11 matrices over $\{0, \ldots, n\}$ whose rows and columns are nondecreasing, and carries a product formula for $a(n)$ marked empirical. Reversing rows and columns turns such a matrix into a plane partition inside a $8 \times 11 \times n$ box, so MacMahon's box theorem applies; the triple product telescopes in its third index and leaves a polynomial in n of degree 88. The entry's own expression, cleared of its denominator, becomes an identity between two polynomials of bounded degree, and is therefore settled by a finite exact computation. The box count itself is classical and is already recorded on this entry; what is proved here is the entry's own formula.

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1 The conjecture

OEIS A140940 is "Number of 8 X 11 matrices with elements in 0..n with each row and each column in nondecreasing order. 8,11,n can be permuted, see formula.". It has offset 0 and begins

$$1, 75582, 1057896060, 4783805983320, 9271015995674160, 9183250377581777952, \ldots$$

The entry states, as conjectures contributed by its author and never marked settled:

> (empirical) set p,q,r to n,11,8 (in any order) in s=p+q+r-1; a(n) = product i in 0..r-1 (binomial(s,p+i)*i!/(s-i)^(r-i-1))

As of the "Last modified" line on the live entry (Jan 25 2025 12:25:52, revision 7) these are still recorded as empirical, and nothing on the entry records them as proved.

2 What is being counted

A matrix x of shape 8×11 with entries in $\{0, \ldots, n\}$, nondecreasing along every row and down every column, becomes, on reversing the order of the rows and of the columns, an array that is NONINCREASING along rows and down columns with the same entries. That is exactly a plane partition whose parts are at most n and whose shape fits inside a 8×11 rectangle — a plane partition in a $8 \times 11 \times n$ box. The correspondence is a bijection, being an involution on arrays, so

$$a(n) \;=\; \mathrm{PP}(8, 11, n).$$

Note which parameter grows: the shape is FIXED and it is the alphabet $0..n$ that n counts, so nothing about transfer matrices applies here.

3 The box count is a polynomial

Theorem 1 (MacMahon)**.**

$$\mathrm{PP}(p,q,r) \;=\; \prod_{i=1}^{p}\prod_{j=1}^{q}\prod_{k=1}^{r}\frac{i+j+k-1}{i+j+k-2}.$$

This is classical; see Stanley [2], and it is already recorded on this entry.

Theorem 2.

$$a(n) \;=\; \prod_{i=1}^{8}\prod_{j=1}^{11}\frac{n+i+j-1}{i+j-1},$$

a polynomial in n of degree 88 with rational coefficients.

Proof. In Theorem 1 the inner product over k telescopes: for fixed i and j,

$$\prod_{k=1}^{n}\frac{i+j+k-1}{i+j+k-2} \;=\; \frac{n+i+j-1}{i+j-1},$$

every other factor cancelling against its neighbour. The remaining double product has $8\cdot 11 = 88$ factors, each linear in n, and the denominators are constants. □

So a is a polynomial, and no appeal to Ehrhart theory or to any general position argument is needed to see it.

4 The entry's expression

The entry writes $s = p+q+r-1$ with $(p,q,r) = (n,11,8)$ and claims

$$a(n) \;=\; \prod_{i=0}^{8-1}\binom{s}{n+i}\frac{i!}{(s-i)^{8-1-i}} \;=\; \frac{N(n)}{D(n)}.$$

Lemma 3. *N and D are polynomials in n of degrees 116 and 28 respectively.*

Proof. With $s = n+11+8-1$ one has $s-(n+i) = 11+8-1-i$, a constant, so

$$\binom{s}{n+i} = \binom{s}{11+8-1-i} = \frac{(s-11-8+2+i)(s-11-8+3+i)\cdots s}{(11+8-1-i)!},$$

a polynomial in n of degree $11+8-1-i$. Multiplying over $i = 0,\ldots,8-1$ gives $\deg N = \sum_{i=0}^{8-1}(11+8-1-i) = 116$. The denominator is $\prod_{i=0}^{8-1}(s-i)^{8-1-i}$, of degree $\sum_{i=0}^{8-1}(8-1-i) = 28$, and it does not vanish for $n \geq 0$. □

Theorem 4. *The entry's expression equals $a(n)$ for every $n \geq 0$.*

Proof. By Theorem 2 and Lemma 3, $a(n)D(n) - N(n)$ is a polynomial in n of degree at most $88+28 = 116$. It was evaluated in exact rational arithmetic at $n = 0,1,\ldots,116$, that is at 117 points, and vanishes at all of them; a nonzero polynomial of degree at most 116 has at most 116 roots, so it is identically zero. Since $D(n) \neq 0$ for $n \geq 0$, the two expressions agree there. □

5 Verification

Three checks, each independent of the algebra above.

First, Theorem 2 reproduces all 7 terms the entry publishes, exactly, in integer arithmetic.

Second, the matrices themselves were counted for the smallest alphabets, directly from the definition: the nondecreasing rows were listed and the chains of 8 of them increasing componentwise counted, with no product formula and no plane partitions anywhere. The counts agree, which is what ties the reversal bijection to the entry's own wording.

Third, the polynomial identity was evaluated over the whole range $n = 0, \ldots, 116$ rather than sampled, in exact rational arithmetic.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A140940.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999. (Plane partitions with bounded part size, Section 7.21.)

[3] P. A. MacMahon, *Combinatory Analysis*, Volume 2, Cambridge University Press, 1916.

[4] G. E. Andrews, *The Theory of Partitions*, Addison–Wesley, 1976.